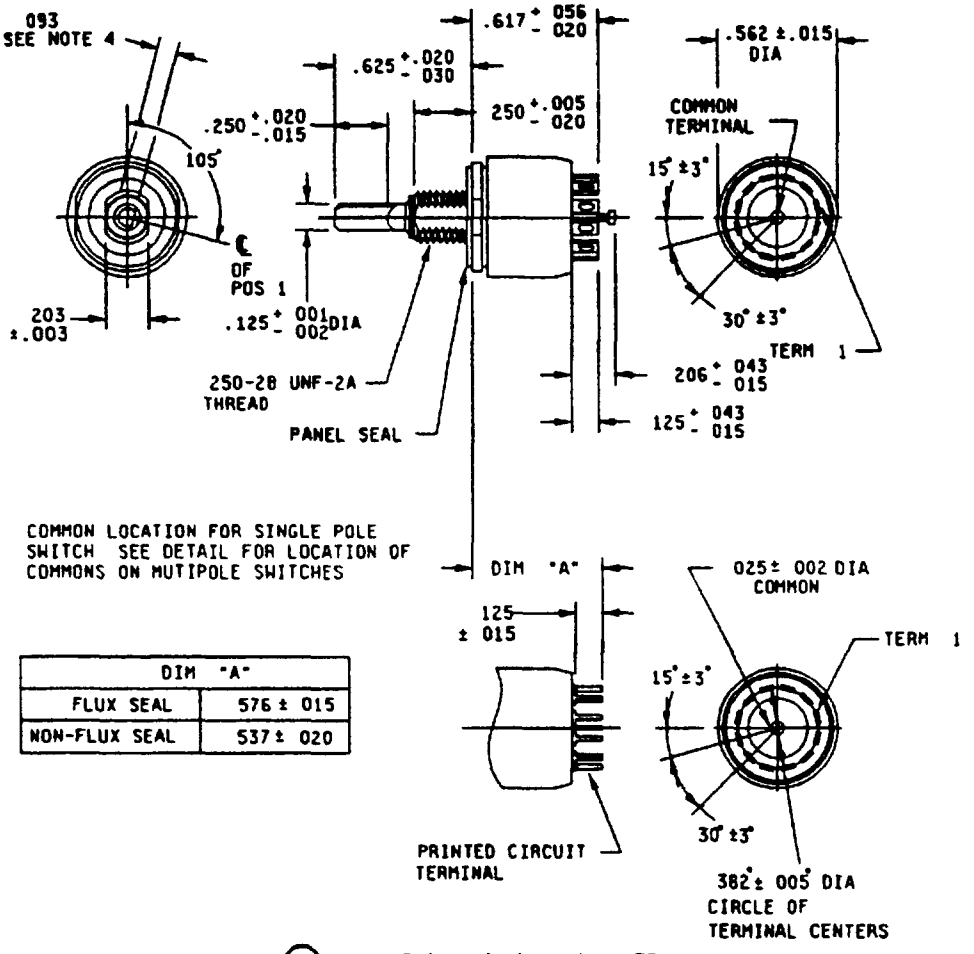
(E) MILITARY SPECIFICATION SHEET

SWITCH, ROTARY, CLOSED CONSTRUCTION, EXPLOSION PROOF, FLUX SEAL AND NON-FLUX SEAL, .200 AMPERE, STYLE SR35

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-3786.



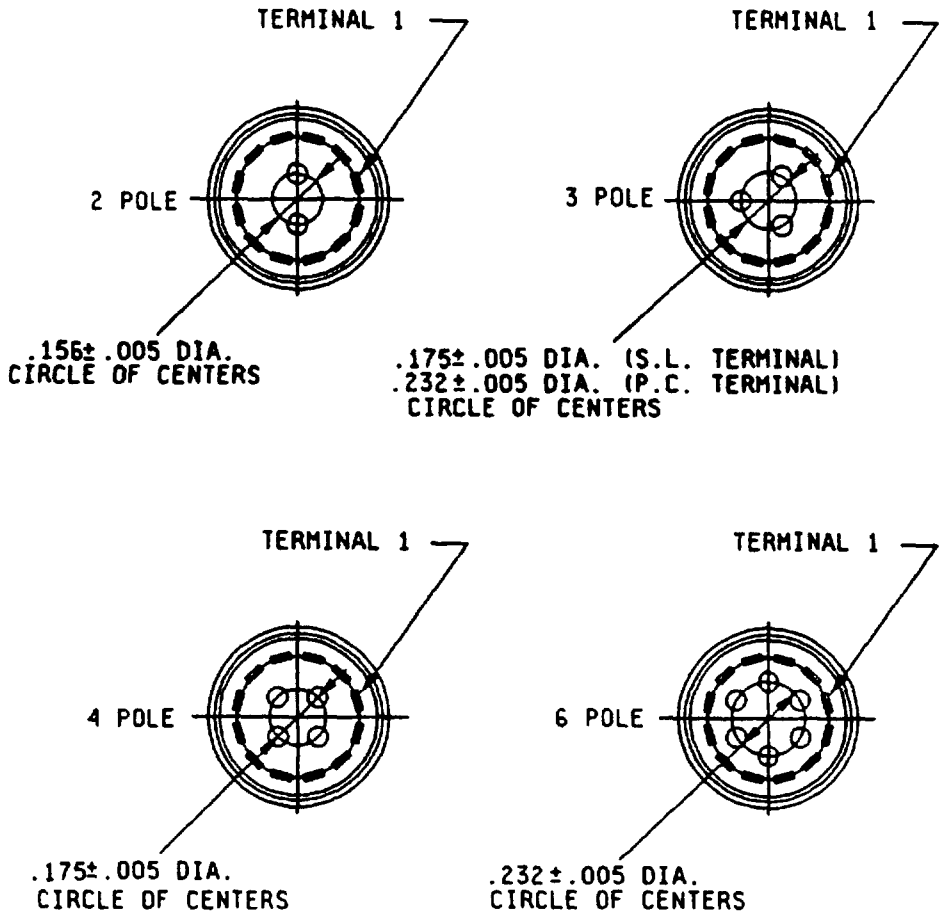
DIM "A"	
FLUX SEAL	576 ± 015
NON-FLUX SEAL	537 ± 020

(E) FIGURE 1. Switch style SR35.

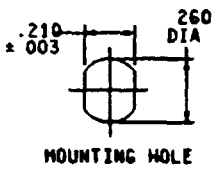
(E) denotes changes

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

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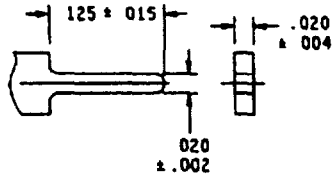
Terminal configurations viewed from rear.



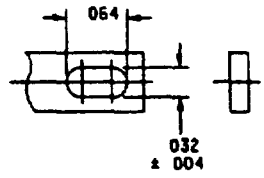
Mounting hole

(E) FIGURE 1. Switch style SR35 - Continued.

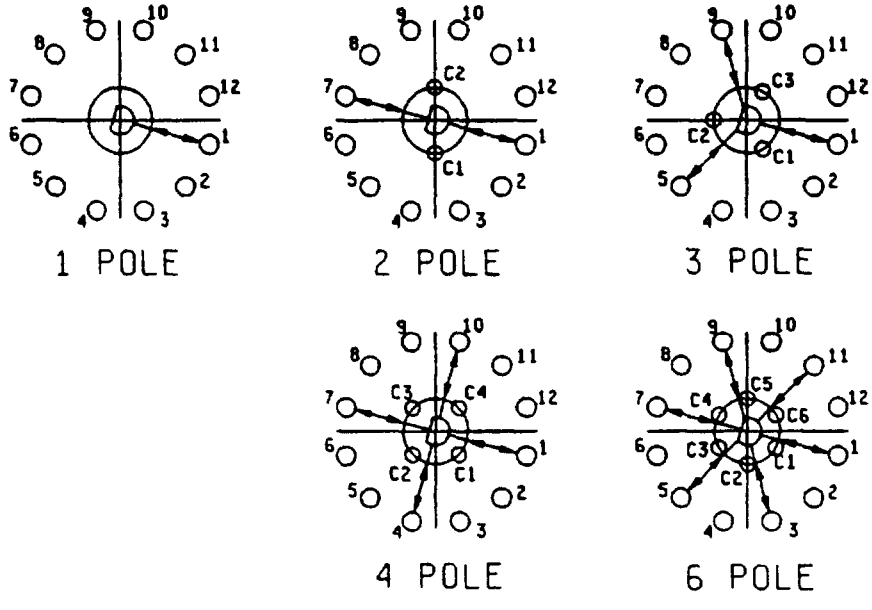
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Printed circuit terminal detail



Solder lug terminal detail



Circuit diagram

Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.001	0.03	.007	0.18	.064	1.63	.187	4.75	.562	14.27
.002	0.05	.015	0.38	.093	2.36	.203	5.16	.576	14.63
.003	0.08	.020	0.51	.125	3.18	.206	5.23	.617	15.67
.004	0.10	.025	0.64	.156	3.96	.250	6.35	.625	15.88
.005	0.13	.032	0.81	.175	4.45	.382	9.70	.667	16.94

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm) and ±3° on angles (noncumulative).
4. Shaft flat is located diametrically opposite to the number one terminal, when in extreme counterclockwise position.

(E) FIGURE 1. Switch style SR35 - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure.

Angle of throw: 30°.

(E) Construction styles: Symbols F, J, and K.

Insulation: Symbol P (plastic).

Mounting hardware: Each switch shall be provided with one corrosion-resistant steel hexagon nut, .089 ±.010-inch thick by .375 inch across the flats and one internal-toothed lock washer, .396/.408 diameter.

Moisture resistance: Insulation resistance measured immediately after conclusion of the moisture resistance test and while the switches are still in the humidity chamber, shall be greater than 100 megohms. At the end of the drying period, the insulation resistance shall be not less than 5,000 megohms.

Temperature-life characteristic: B (25,000 cycles) (-65°C and +85°C).

Vibration grade: 3 (10 to 2,000 Hz).

(E) Shock symbol: B (high and medium impact)

Altitude: C (70,000 ft.).

Terminal strength (pull): A force of 1.5 pounds shall be applied to the terminals.

Terminal markings: Switches may be marked with terminal numbers on side of switch.

Stop strength: Stops shall withstand a force of 10 pound-inches.

(E) Flux seal: Applicable when specified in Part or Identifying Number (PIN).

Rotational torque: The torque for shaft rotation shall be within the limits specified in table I.

TABLE I. Rotational-torque limits.

Temperature	Torque (lb-in)	
	Minimum	Maximum
Room	.5	1.5
Minimum	.5	2.0

Life (rotational): The test loads for the applicable circuit conditions shall be as specified in table II. Each of the loads, specified for the applicable environmental condition, shall be switched by at least one rotor contact of the switch.

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(E) TABLE II. Electrical loads.

Environmental condition	Lamp load (tungsten)		Inductive load (2.8 henries)		Resistive load	
	Milliamperes	Volts dc	Milliamperes	Volts dc	Milliamperes	Volts
At atmospheric pressure	100	28	30	28	200	28 dc
					75	115 V rms 60 Hz
At reduced barometric pressure	---	---	---	---	200	28 dc
					75	115 V rms 60 Hz

Dielectric withstanding voltage: The magnitude of test voltage for the dielectric withstanding voltage shall be 600 volts, alternating current (ac) atmospheric pressure, and 250 volts ac at reduced barometric pressure.

Test procedure on lamp load at atmospheric pressure:

The test procedure on lamp load at atmospheric pressure shall be as follows:

a. Test potential and load. One pole on each of eight switches to be tested, four at atmospheric and four at reduced barometric pressure, shall be energized by the specified lamp load. A common terminal and one selected terminal shall be arranged to be "on" once per cycle. Adjacent terminals shall not be energized. Suitable means shall be provided to indicate when contacts have failed to make and break.

b. Cycling and cycle rate. Ten cycles per minute. At the option of the supplier, cycling may be interrupted, periodically, after each 5,000 cycles of operation.

Ordering data: Acquisition documents should specify the following:

a. Title, number, and date of this specification sheet, and the military Part or Identifying Number (PIN) as listed in table III, example A, or tables IV, V, and VI, example B.

b. For switches not listed in table III, acquisition shall be in accordance with the ordering data of MIL-S-3786 (for switches covered by specification sheets).

(E) Military PIN: The PIN shall be assigned by using example A, which consists of a dash number from table III. If the desired characteristics are not defined, the PIN shall be assigned by using the code letters from tables IV, V, and VI as described in example B illustrated below:

Example A: Non-flux sealed switches:

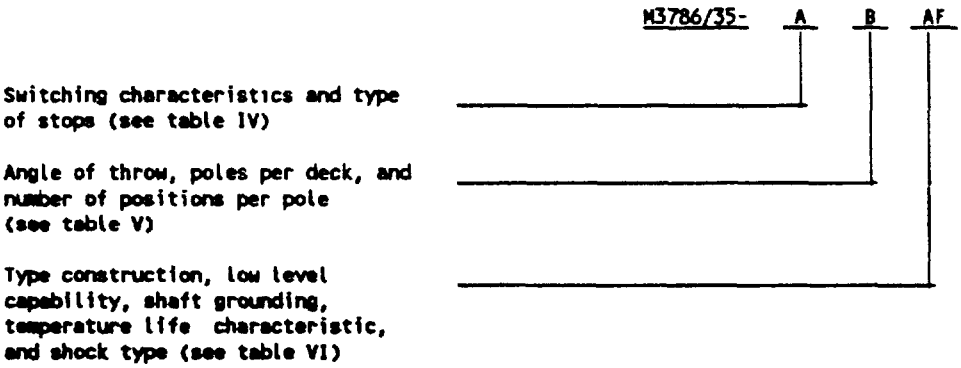
M3786/35- 001

Switching characteristics (see table III)

PIN M3786/35-001 identifies construction style F, explosion proof rotary switch with a 30° angle of throw, 1 pole, 2 through 12 positions per pole, adjustable stops, solder lug termination with nonshorting switching characteristics.

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Example B: Flux sealed switches:



PIN M3786/35-ABAF identifies a rotary switch with fixed stops, nonshorting switching characteristics, and pc terminals, that is explosion proof, flux proof, closed construction, sealed shaft and panel seal, 30° angle of throw, 1 pole per deck, and 7 positions per pole.

(E) TABLE III. Switch characteristics and PIN's for type F construction.

PIN M3786/35-	Number of poles	Number of positions per pole	Type of stops	Termination	Switching characteristic (S or NS)
001	1	2 through 12	Adjustable	Solder lug	NS
002	2	2 through 6	"	Solder lug	"
003	4	2 through 3	"	Solder lug	"
004	1	2 through 12	"	Printed circuit	"
005	2	2 through 6	"	Printed circuit	"
006	4	2 through 3	"	Printed circuit	"
007	1	2	Fixed	Solder lug	"
008	1	3	"	"	"
009	1	4	"	"	"
010	1	5	"	"	"
011	1	6	"	"	"
012	1	7	"	"	"
013	1	8	"	"	"
014	1	9	"	"	"
015	1	10	"	"	"
016	1	11	"	"	"
017	1	12	"	"	"
018	2	2	"	"	"
019	2	3	"	"	"
020	2	4	"	"	"
021	2	5	"	"	"
022	2	6	"	"	"
023	4	2	"	"	"
024	4	3	"	"	"
025	1	2	"	Printed circuit	"
026	1	3	"	"	"
027	1	4	"	"	"
028	1	5	"	"	"
029	1	6	"	"	"

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(E) TABLE III. Switch characteristics and PIN's for type F construction - Continued.

PIN M3786/35-	Number of poles	Number of positions per pole	Type of stops	Termination	Switching characteristic (S or NS)
030	1	7	Fixed	Printed circuit	NS
031	1	8	"	"	"
032	1	9	"	"	"
033	1	10	"	"	"
034	1	11	"	"	"
035	1	12	"	"	"
036	2	2	"	"	"
037	2	3	"	"	"
038	2	4	"	"	"
039	2	5	"	"	"
040	2	6	"	"	"
041	4	2	"	"	"
042	4	3	"	"	"
043	1	2	"	Solder lug	S
044	1	3	"	"	"
045	1	4	"	"	"
046	1	5	"	"	"
047	1	6	"	"	"
048	1	7	"	"	"
049	1	8	"	"	"
050	1	9	"	"	"
051	1	10	"	"	"
052	1	11	"	"	"
053	1	12	"	"	"
054	2	2	"	"	"
055	2	3	"	"	"
056	2	4	"	"	"
057	2	5	"	"	"
058	2	6	"	"	"
059	4	2	"	"	"
060	4	3	"	"	"
061	1	2	"	Printed circuit	"
062	1	3	"	"	"
063	1	4	"	"	"
064	1	5	"	"	"
065	1	6	"	"	"
066	1	7	"	"	"
067	1	8	"	"	"
068	1	9	"	"	"
069	1	10	"	"	"
070	1	11	"	"	"
071	1	12	"	"	"
072	2	2	"	"	"
073	2	3	"	"	"
074	2	4	"	"	"
075	2	5	"	"	"
076	2	6	"	"	"
077	4	2	"	"	"
078	4	3	"	"	"

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(E) TABLE III. Switch characteristics and PIN's for type F construction - Continued.

PIN M3786/35-	Number of poles	Number of positions per pole	Type of stops	Termination	Switching characteristic (S or NS)
079 1/	1	12	None	Solder lug	NS
080 1/	1	12	"	Printed circuit	NS
081 1/	1	12	"	Solder lug	S
082 1/	1	12	"	Printed circuit	S
083	3	2	Fixed	Solder lug	NS
084	3	3	"	"	"
085	3	4	"	"	"
086	6	2	"	"	"
087	3	2	"	Printed circuit	"
088	3	3	"	"	"
089	3	4	"	"	"
090	6	2	"	"	"
091	3	2	"	Solder lug	S
092	3	3	"	"	"
093	3	4	"	"	"
094	6	2	"	"	"
095	3	2	"	Printed circuit	"
096	3	3	"	"	"
097	3	4	"	"	"
098	6	2	"	"	"

1/ Continuous rotation.

(E) TABLE IV. Switching characteristics.

Code letter	Switching characteristics	Type of stops
A	Nonshorting	Fixed
B	Shorting	Adjustable
C	Nonshorting	Fixed
D	Shorting	Adjustable

(E) TABLE V. Code letter for type construction, low level capability, shaft grounding, and type of termination.

Code letter	Type construction	PC terminals	Solder lug terminals
A	J	Yes	No
B	K	Yes	No
C	J	No	Yes
D	K	No	Yes

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(E) TABLE VI. Code letter for combination of angle of throw, poles per deck, and number of positions per pole.

Code letter	Angle of throw	Poles per deck	Number of pos	Code letter	Angle of throw	Poles per deck	Number of pos
AA	30	1	2	AN	30	2	2
AB	30	1	3	AP	30	2	3
AC	30	1	4	AR	30	2	4
AD	30	1	5	AT	30	2	5
AE	30	1	6	AV	30	2	6
AF	30	1	7	AW	30	3	2
AG	30	1	8	AY	30	3	3
AH	30	1	9	BA	30	3	4
AJ	30	1	10	BB	30	4	2
AK	30	1	11	BC	30	4	3
AL	30	1	12	BD	30	6	2
AM	30	1	12 1/				

1/ Continuous rotation.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, AV, ME, MI, SM
Navy - AS, CG, MC, OS
Air Force - 19
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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